

#1 User: I cooked with sunflower oil once, it was delicious and mild-flavored.

#2 User: I'm allergic to peanuts. Please avoid peanut oil.

#4 User: Yesterday I ate satay and got hives.

#3 Assistant: Got it, no peanuts, no peanut oil.

Query: What oil can I use instead of peanut oil?

(a) Frequent Rewriting Memory

Original #1: Sunflower oil was delicious and mild-flavored.

Add #2; Rewrite #1: peanut allergy; avoid peanut oils; prefer mild-flavored oil.

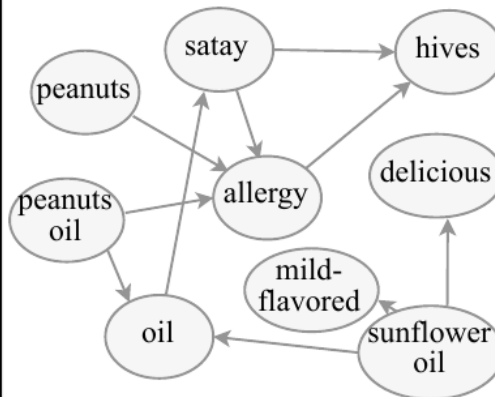
Add #3; Rewrite #1: Nut allergy; avoid nut-based oil; prefer mild oils.

Add #4; Rewrite #1: Had hives after satay; avoid oily foods; prefers mild oils.



Answer: You should avoid oily foods, and can try **mild oils**.

(b) Graph-based Indexing

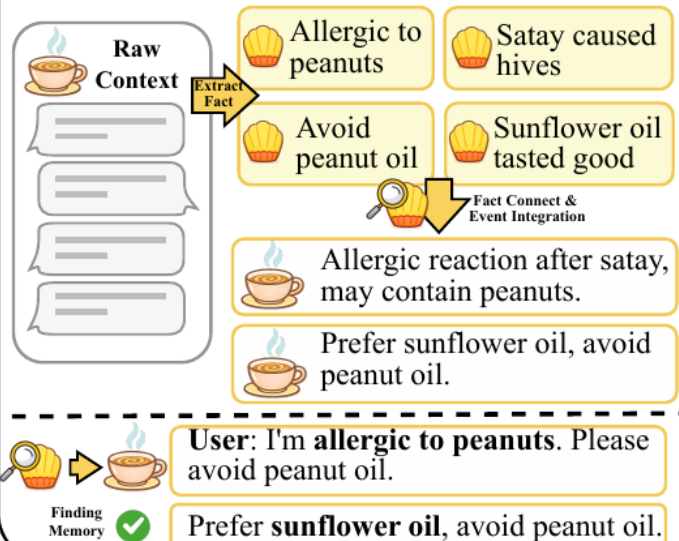


Graph Traversal:

peanut oil → allergy → **hives**
peanut oil → oil → satay → **hives**



(c) AnchorMem (Ours)



ACL 2026 Findings



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AnchorMem: Anchored Facts with Associative Contexts for Building Memory in Large Language Models